CONSTANTS

$M_e = 5.98 \cdot 10^{24}$ kg

$R_e = 6.37 \cdot 10^{6}$ m

$g = 9.8$ m/s^2

$G = 6.67 \cdot 10^{-11}$ N·m^2/kg^2

$v_{\text{sound}} = 343$ m/s

$m_p = 1.67 \cdot 10^{-27}$ kg

$m_e = 9.11 \cdot 10^{-31}$ kg

$e = 1.60 \cdot 10^{-19}$ C

$k_B = 1.38 \cdot 10^{-23}$ J/K

$R = 8.31$ J/mol-K

$N_A = 6.02 \cdot 10^{23}$ particles/mol

$k = 1/4\pi\epsilon_0 = 8.99 \cdot 10^{9}$ N·m^2/C^2

$\epsilon_0 = 8.85 \cdot 10^{-12}$ C^2/N·m^2

$\mu_0 = 4\pi \cdot 10^{-7}$ T·m/A

$c = 3.00 \cdot 10^{8}$ m/s

$h = 6.63 \cdot 10^{-34}$ J·s

Conversions

$T(°\text{C}) = T(\text{K}) - 273.15$

$1\ \text{m}^3 = 1000\ \text{L}$

1 atm = 101,300 Pa = 760 mm Hg = 14.7 psi

$1\ \text{eV} = 1.60 \cdot 10^{-19}$ J

Basic Formulas

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

KINEMATICS

$\vec{v} = \frac{d\vec{x}}{dt}$ (slope of pos.) $\quad \vec{a} = \frac{d\vec{v}}{dt}$ (slope of vel.)

Circular Motion

$d = \theta R \quad v = \omega R \quad a = \alpha R$

$f = \frac{1}{T} \quad \omega = 2\pi f$

$a_c = v^2/R$

FORCE AND ENERGY

	Force	**Energy**
Gravity (on earth)	$w = mg$	$U_g = mgh$
Gravity (in space)	$F_g = \frac{Gm_1m_2}{r^2}$	$U_g = -\frac{Gm_1m_2}{r}$
Spring	$F_{sp} = k\Delta x$	$U_{sp} = \frac{1}{2}k(\Delta x)^2$
External Force	F_{ext}	$W = F_{\text{ext}}d\cos\theta$
Kinetic Friction	$f_k = \mu_k N$	$W = -f_k d$
Drag	D	
Buoyancy	$F_B = \rho_f V_f g$	
Electric	$F_{\text{el}} = qE$	$U_{\text{el}} = qV$
Magnetic	$F_{\text{mag}} = \lvert q\rvert vB\sin\theta$	$W = 0$
Magnetic	$F_{\text{mag-wire}} = ILB$	$W = 0$
Normal Force	$N \le N_{\text{max}}$	$W = 0$ (for ground)
Tension	$T \le T_{\text{max}}$	$W = Td\cos\theta$
Static Friction	$f_s \le \mu_s N$	$W = 0$ (for ground)
Kinetic Energy		$K = \frac{1}{2}mv^2$
Rotational Kinetic		$K_{\text{rot}} = \frac{1}{2}I\omega^2$
Chemical Energy		E_{chem}
Thermal Energy		E_{th}

Newton's Law

$$\sum \vec{F}_{\text{ext}} = m\vec{a} \qquad \sum \tau_{\text{ext}} = I\alpha$$

ROTATIONAL MOTION

$$x_{\text{cm}} = \frac{x_1 m_1 + x_2 m_2 + \cdots}{m_{\text{tot}}} \qquad y_{\text{cm}} = \frac{y_1 m_1 + y_2 m_2 + \cdots}{m_{\text{tot}}}$$

$$\tau = rF_{\perp} = r_{\perp}F = rF\sin\phi$$

Moments of Inertia

$$I_{\text{pt}} = I_{\text{ring}} = MR^2 \qquad I_{\text{rod ab end}} = \frac{1}{3}ML^2 \qquad I_{\text{rod ab mid}} = \frac{1}{12}ML^2$$

$$I_{\text{disk}} = \frac{1}{2}MR^2 \qquad I_{\text{solid sph}} = \frac{2}{5}MR^2 \qquad I_{\text{sph shell}} = \frac{2}{3}MR^2$$

Rolling

$$v_{\text{cm}} = \omega R$$

$$a_{\text{cm}} = \alpha R$$

MOMENTUM AND ENERGY

$$\vec{p} = m\vec{v}$$

$$L = I\omega$$

$$\vec{J} = \vec{F}_{\text{avg}}\Delta t$$

$$J_{\text{rot}} = \tau_{\text{avg}}\Delta t$$

$$P = \frac{\Delta E}{\Delta t} \text{ or } P = \frac{W}{\Delta t} = Fv$$

Conservation Laws

$$\vec{p}_i + \sum \vec{J}_{\text{ext}} = \vec{p}_f$$

$$L_i + \sum J_{\text{rot ext}} = L_f$$

$$E_i + \sum W_{\text{ext}} + \sum Q_{\text{ext}} = E_f$$

FLUIDS

$$\rho = \frac{m}{V} \qquad P = \frac{F}{A}$$

$$Q = vA \qquad v_1 A_1 = v_2 A_2$$

$$P = P_0 + \rho g h \quad \text{(fluid not moving)}$$

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$$

$$Q = \frac{\pi R^4 \Delta P}{8\eta L}$$

OSCILLATIONS AND WAVES

General Oscillations

$$f = \frac{1}{T} \qquad \omega = 2\pi f = \frac{2\pi}{T}$$

$$x(t) = A\cos(\omega t) \quad v(t) = -\omega A\sin(\omega t) \quad a(t) = -\omega^2 A\cos(\omega t)$$

General Waves

$$f = \frac{1}{T} \qquad \omega = 2\pi f = \frac{2\pi}{T} \qquad k = \frac{2\pi}{\lambda} \qquad v = f\lambda$$

$$y(x,t) = A\cos(kx - \omega t)$$

$$I = \frac{P}{A} = \frac{P_{\text{source}}}{4\pi r^2}$$

Specific Examples

$$\omega_{\text{mass-spring}} = \sqrt{k/m}$$

$$\omega_{\text{pendulum}} = \sqrt{g/L} \qquad \omega_{\text{phys pendulum}} = \sqrt{mgd/I}$$

$$v_{\text{string}} = \sqrt{T/\mu}$$